

ASSIGNMENT : 10 – IP ADDRESSING

GUIDED BY : VIPUL MISTRI

Q.1 Find the IP address of your computer or mobile device and write down whether it is IPv4 or IPv6. Also mention if it is a public or private IP.

ANS :

```
C:\Windows\system32\cmd.exe

C:\Users\Admin>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet0:

    Connection-specific DNS Suffix  . : localdomain
    Link-local IPv6 Address . . . . . : fe80::b5d5:93a9:6618:44f1%6
    IPv4 Address. . . . . : 192.168.31.131
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.31.2

C:\Users\Admin>
```

This is a ipv4 address and it is a private ip address.

Q.2 Given the following list of IP addresses, identify which ones are valid private IPs and which are public: 192.168.1.10, 10.0.0.5, 172.16.5.20, 8.8.8.8, 123.45.67.89. Hint: Check the standard private IP address ranges for IPv4.

ANS : Here is the breakdown of the IP addresses based on the standard IPv4 private address ranges defined in RFC 1918:

- **Private IP Address Ranges:**
 - **Class A:** 10.0.0.0 to 10.255.255.255
 - **Class B:** 172.16.0.0 to 172.31.255.255
 - **Class C:** 192.168.0.0 to 192.168.255.255

Classification

- 192.168.1.10 — **Private** (Falls within the Class C 192.168.0.0/16 range, commonly used in local home networks).
- 10.0.0.5 — **Private** (Falls within the Class A 10.0.0.0/8 range, used in private internal networks).
- 172.16.5.20 — **Private** (Falls within the Class B 172.16.0.0/12 range, used in private corporate/lab environments).

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- 8.8.8.8 — **Public** (Globally routable IP address belonging to Google's public DNS server).
- 123.45.67.89 — **Public** (Globally routable IP address used on the public internet).

Q.3 Write a short script in Python or JavaScript that takes an IP address as input and tells the user whether it is a valid IPv4 address or not. Constraint: Do not use any third-party libraries for validation.

ANS : Python

```
def is_valid_ipv4(ip: str) -> bool:
```

```
    parts = ip.split(".")
```

```
    if len(parts) != 4:
```

```
        return False
```

```
    for part in parts:
```

```
        if not part.isdigit() or (len(part) > 1 and part.startswith("0")):
```

```
            return False
```

```
        if not (0 <= int(part) <= 255):
```

```
            return False
```

```
    return True
```

```
# Test
```

```
ip = input("Enter IP: ")
```

```
print("Valid IPv4" if is_valid_ipv4(ip) else "Invalid IPv4")
```

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Q.4 Explain with an example how Instagram or WhatsApp might use IP addresses to detect suspicious login activity from different locations.

ANS : Here is how Instagram or WhatsApp uses IP addresses to detect suspicious login attempts:

1. IP Geolocation Lookup

Every public IP address is registered to an Internet Service Provider (ISP) and tied to a geographic region (city, state, country). When you log in, the app extracts your client's IP address and maps it to a physical location.

2. Baseline User Profile

The platform maintains a record of your typical access patterns:

- **Known Devices & IPs:** Your home Wi-Fi IP (e.g., Surat, India) or mobile carrier IP.
- **Known Locations:** Cities or regions you frequently log in from.

Real-World Example: Impossible Travel Scenario

1. **8:00 AM — Normal Login:** You log into Instagram on your phone connected to your home Wi-Fi in **Surat, India** (IP: 103.21.124.5). Instagram records your active session in Surat.
2. **8:15 AM — Suspicious Attempt:** 15 minutes later, a login request for your account comes from an IP address located in **Frankfurt, Germany** (IP: 185.220.101.5).

How the Security System Reacts:

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- **Travel Time Verification:** The security algorithm calculates that traveling between Surat and Frankfurt in 15 minutes is physically impossible.
- **Risk Assessment:** The system flags the Germany IP as a high-risk anomaly (often indicating a compromised password or the use of a VPN/Proxy).
- **Automated Action:**
 - Blocks the incoming session from Germany.
 - Sends an immediate push notification or email: *"Unusual login attempt detected near Frankfurt, Germany."*
 - Requires Two-Factor Authentication (2FA) or password reset before allowing access.